Unit- 1

AI &ML

What is Artificial Intelligence?

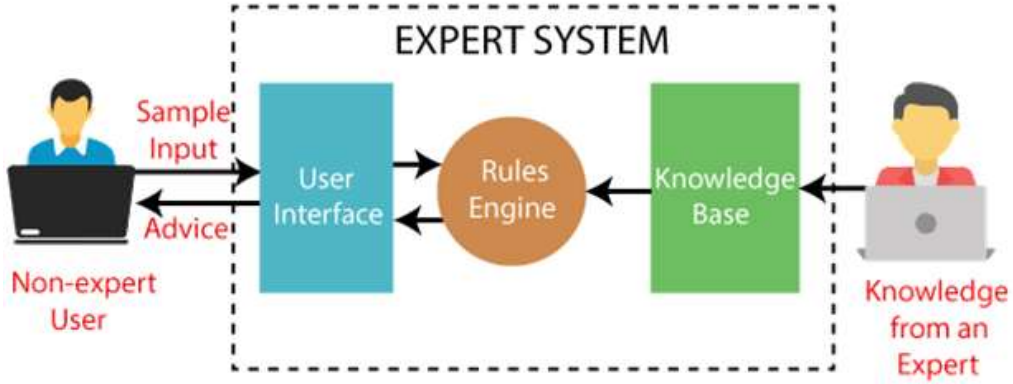
- According to the father of Artificial Intelligence John McCarthy, it is

"The science and engineering of making intelligent machines, especially intelligent computer programs".

- Artificial Intelligence is a way of **making a computer, a computer-controlled robot, or a software think intelligently**, in the similar manner the intelligent humans think.
- AI is accomplished by studying how human brain thinks, and how humans learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems.

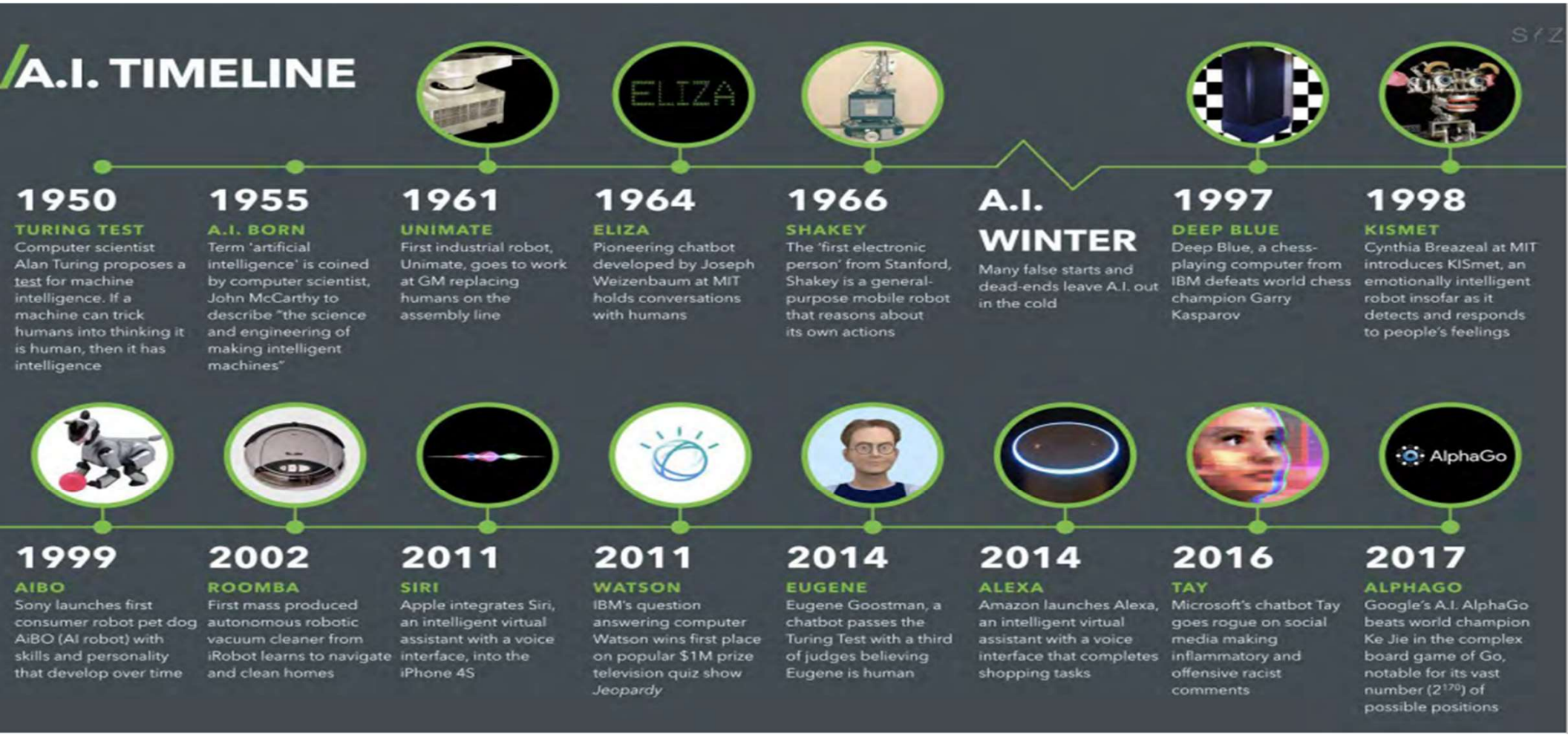
Goals of AI

- **To Create Expert Systems:** The systems which exhibit intelligent behavior, learn, demonstrate, explain, and advice its users.

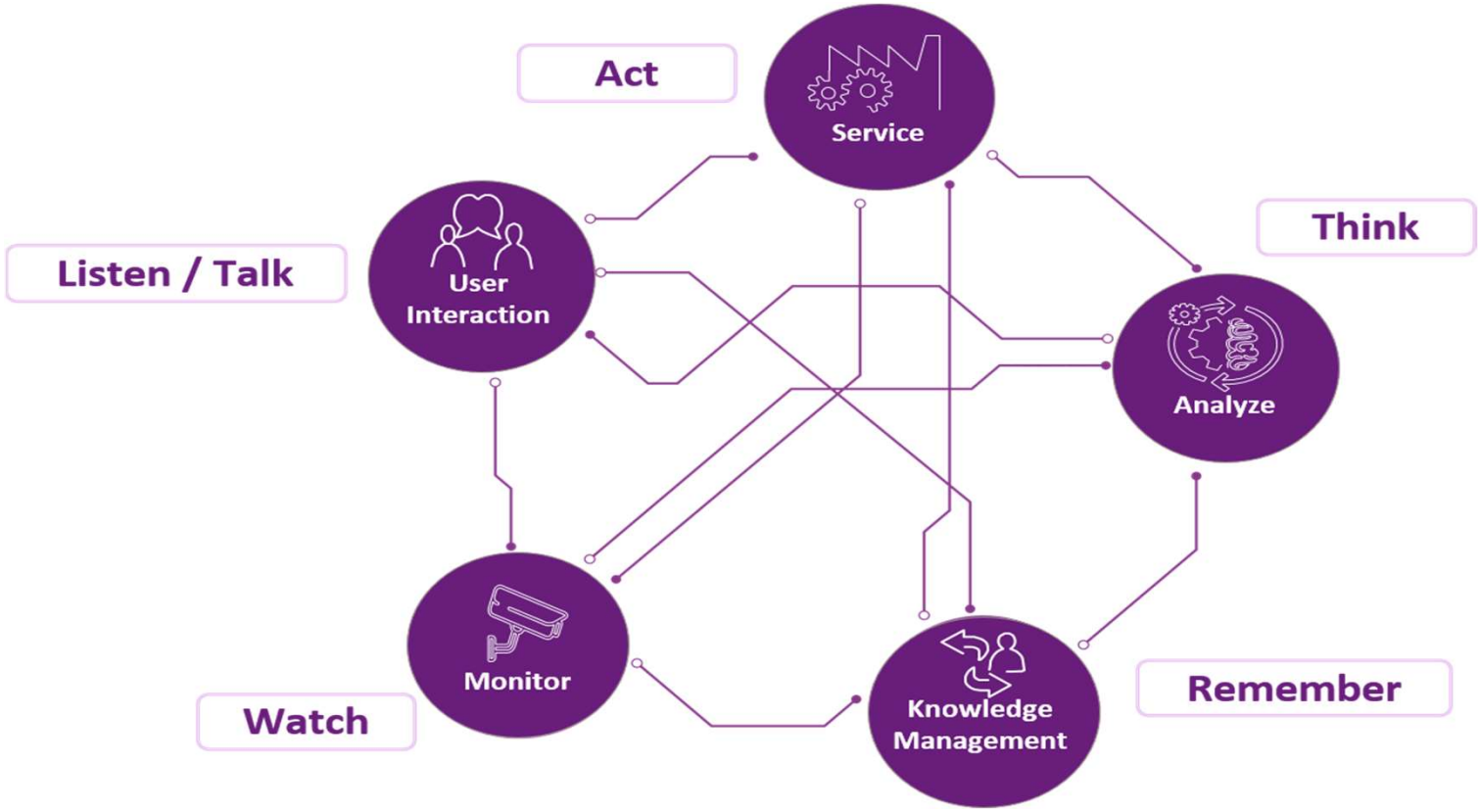


- **To Implement Human Intelligence in Machines:-** Creating systems that understand, think, learn, and behave like humans.

History of AI



Five senses of Artificial Intelligence



- **Interaction (talk/listen)—**This is the ability to listen, read, talk, write, and respond to users of the AI solution. The aim here is for technology to ensure that the interaction feels intuitive and the customer is happy. Examples in this space include chatbots and voice bots.

- **Monitor (watch)—**Here technology is used to watch and record key business data. It is used to create knowledge. This would include CCTV and IoT sensors.

- **Knowledge (remember)—**This is about being able to store and find information effectively using components like databases and search engines. This is probably the worst-developed area within corporations, but examples include Wikipedia and my hard drive.

- **Analyze (think)—**This is the ability to detect patterns and recognize trends. It applies algorithms to knowledge to determine appropriate action or predict future consequences.

- **Service (act)—**This area uses technology to do things. We are used to the concept of robots working on an assembly line and now they are moving into the office. Examples include resetting a password and placing a customer order.

Sub Areas of AI:

1) **Game Playing:-** Deep Blue Chess program beat world champion Gary Kasparov

2) **Speech Recognition :-**PEGASUS spoken language interface to American Airlines' EAASY SABRE reseration system, which allows users to obtain flight information and make reservations over the telephone. The 1990s has seen significant advances in speech recognition so that limited systems are now successful.

3) **Computer Vision :-** Face recognition programs in use by banks, government, etc. The ALVINN system from CMU autonomously drove a van from Washington, D.C. to San Diego (all but 52 of 2,849 miles), averaging 63 mph day and night, and in all weather conditions. Handwriting recognition, electronics and manufacturing inspection, photo interpretation, baggage inspection, reverse engineering to automatically construct a 3D geometric model.

4) **Expert Systems:-** Application-specific systems that rely on obtaining the knowledge of human experts in an area and programming that knowledge into a system. For example:- Flight-tracking systems, Clinical systems

5) Mathematical Theorem Proving:- Use inference methods to prove new theorems.

6) Natural Language processing :- Understanding AltaVista's translation of web pages. Translation of Caterpillar Truck manuals into 20 languages. For example:- Google Now feature, speech recognition, Automatic voice output

7) Scheduling and Planning:- Automatic scheduling for manufacturing. DARPA's DART system used in Desert Storm and Desert Shield operations to plan logistics of people and supplies. American Airlines rerouting contingency planner. European space agency planning and scheduling of spacecraft assembly, integration and verification.

8) Artificial Neural Networks: Pattern recognition systems such as face recognition, character recognition, handwriting recognition.

Building AI Systems:

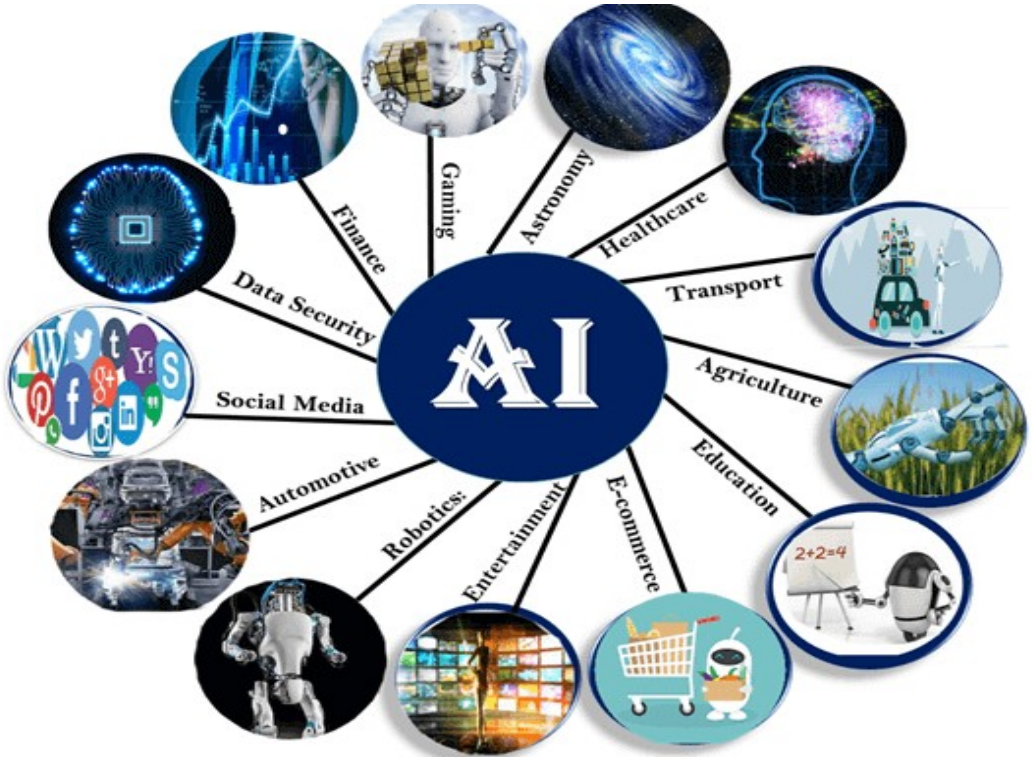
1. **Reasoning :-** It is the set of processes that enables us to provide basis for judgement, making decisions, and prediction.

2. **Learning:-** It is the activity of gaining knowledge or skill by studying, practicing, being taught, or experiencing something. Learning enhances the awareness of the subjects of the study.

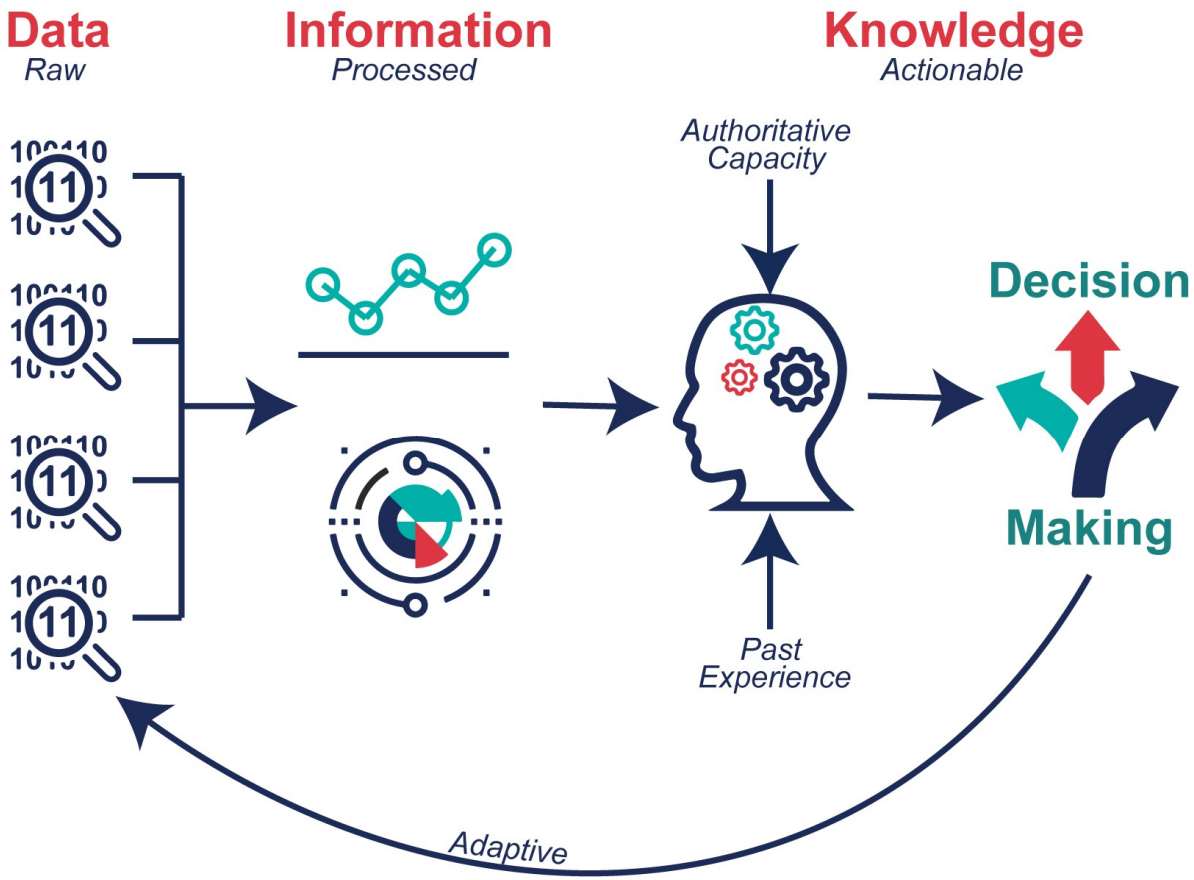
3. **Problem Solving:-** It is the process in which one perceives and tries to arrive at a desired solution from a present situation by taking some path, which is blocked by known or unknown hurdles. Problem solving also includes decision making, which is the process of selecting the best suitable alternative out of multiple alternatives to reach the desired goal are available.

4. **Perception:-**Perception presumes sensing. In humans, perception is aided by sensory organs. In the domain of AI, perception mechanism puts the data acquired by the sensors together in a meaningful manner.

Application of AI



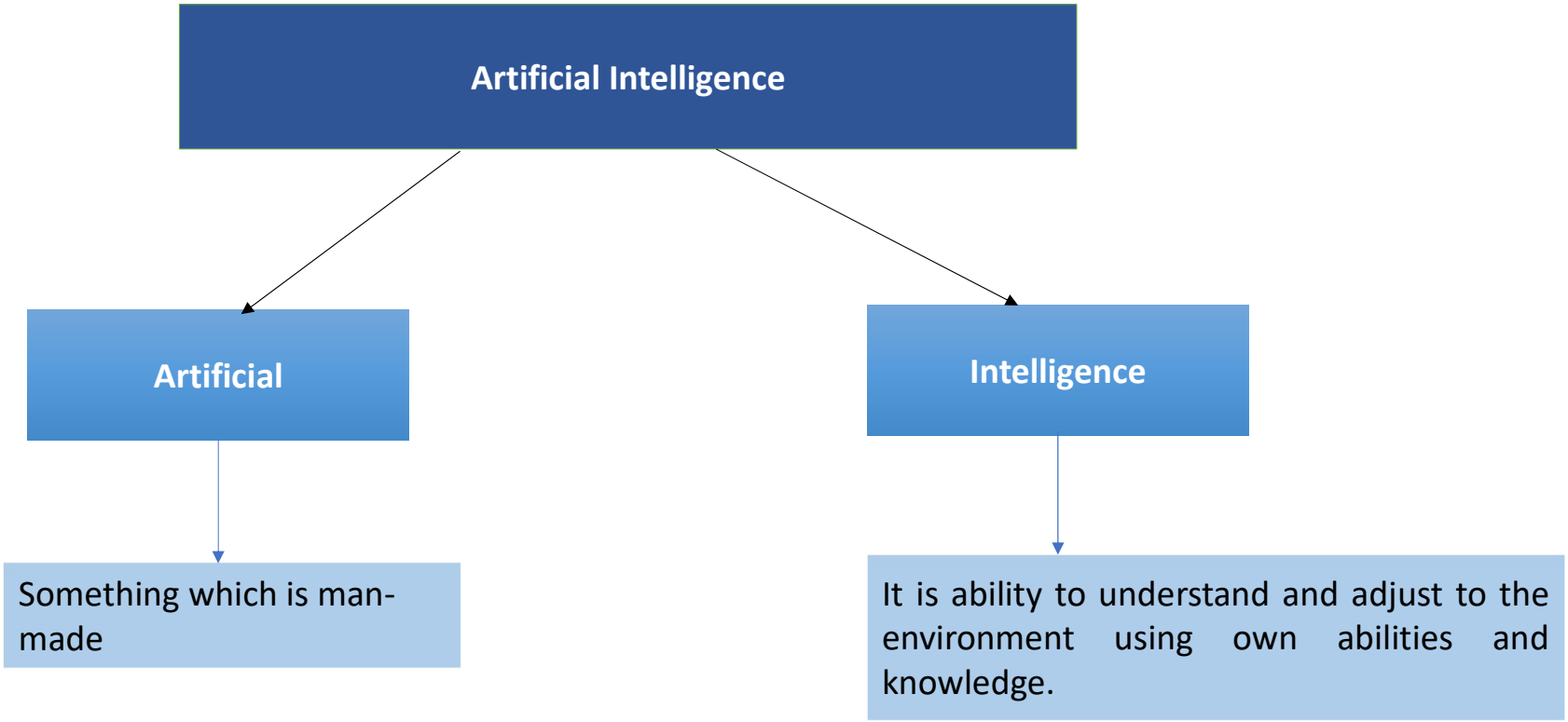
Data, Information, Knowledge



1. **Data** are the raw alphanumeric values obtained through different acquisition methods. Data in their simplest form consist of raw alphanumeric values.

2. **Information** is created when data are processed, organized, or structured to provide context and meaning. Information is essentially processed data.

3. **Knowledge** is what we know. Knowledge is unique to each individual and is the accumulation of past experience and insight that shapes the lens by which we interpret, and assign meaning to, information. For knowledge to result in action, an individual must have the authority and capacity to make and implement a decision. Knowledge (and authority) are needed to produce actionable information that can lead to impact.

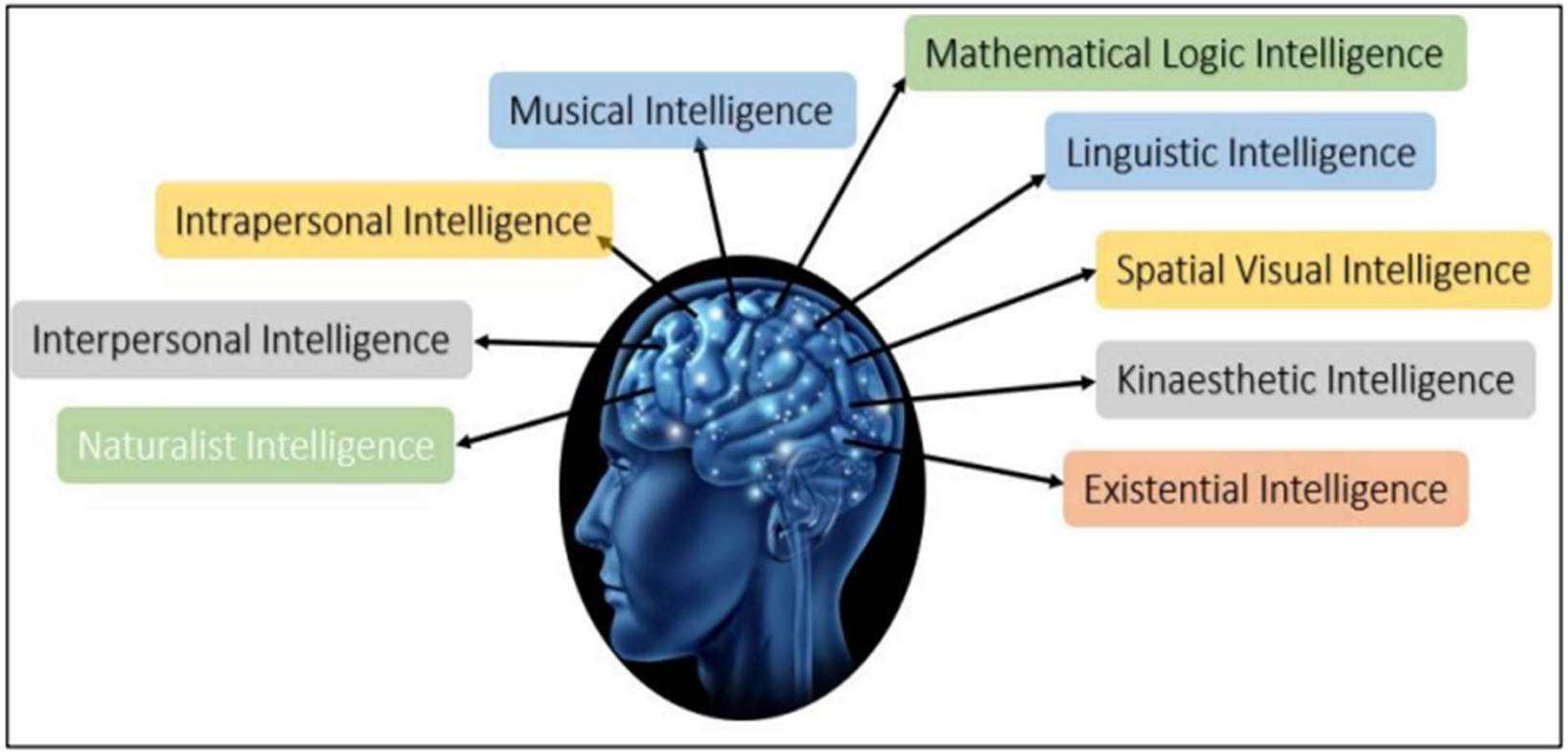

Artificial Intelligence
Artificial
Intelligence
Something which is man-made
It is ability to understand and adjust to the environment using own abilities and knowledge.

Intelligence

1.The capacity for logic, understanding, self-awareness, learning, emotional knowledge, reasoning, planning, creativity, critical thinking, and problem-solving.

2.More generally, it can be described as the **ability to perceive or infer** information and to retain it as knowledge to be applied towards adaptive behaviors within an environment or context.

Traits of Intelligence



Mathematical Logical Reasoning	•A person's ability to regulate, measure, and understand numerical symbols, abstraction and logic.
Linguistic Intelligence	•Language processing skills both in terms of understanding or implementation in writing or verbally.
Spatial Visual Intelligence	•It is defined as the ability to perceive the visual world and the relationship of one object to another.
Kineasthetic Intelligence	•Ability that is related to how a person uses his limbs in a skilled manilr.
Musical Intelligence	•As the name suggests, this intelligence is about a person's ability to recognize and create sounds, rhythms, and sound patterns.
Intrapersonal Intelligence	•Describes how high the level of self-awareness someone has is. Starting from realizing weakness, strength, to his own feelings.
Existential Intelligence	•An additional category of intelligence relating to religious and spiritual awareness.
Naturalist Intelligence	•An additional category of intelligence relating to the ability to process information on the environment around us.
Interpersonal intelligence	•Interpersonal intelligence is the ability to communicate with others by understanding other people's feelings & influence of the person.

INTELLIGENCE

Ability to interact with te real world

To percieve, understand and act

Ablitity to take action: to have an effect

Reasoning and Planning

Modelling the external world: given input

Problem Solving, Planning and Decision

Making

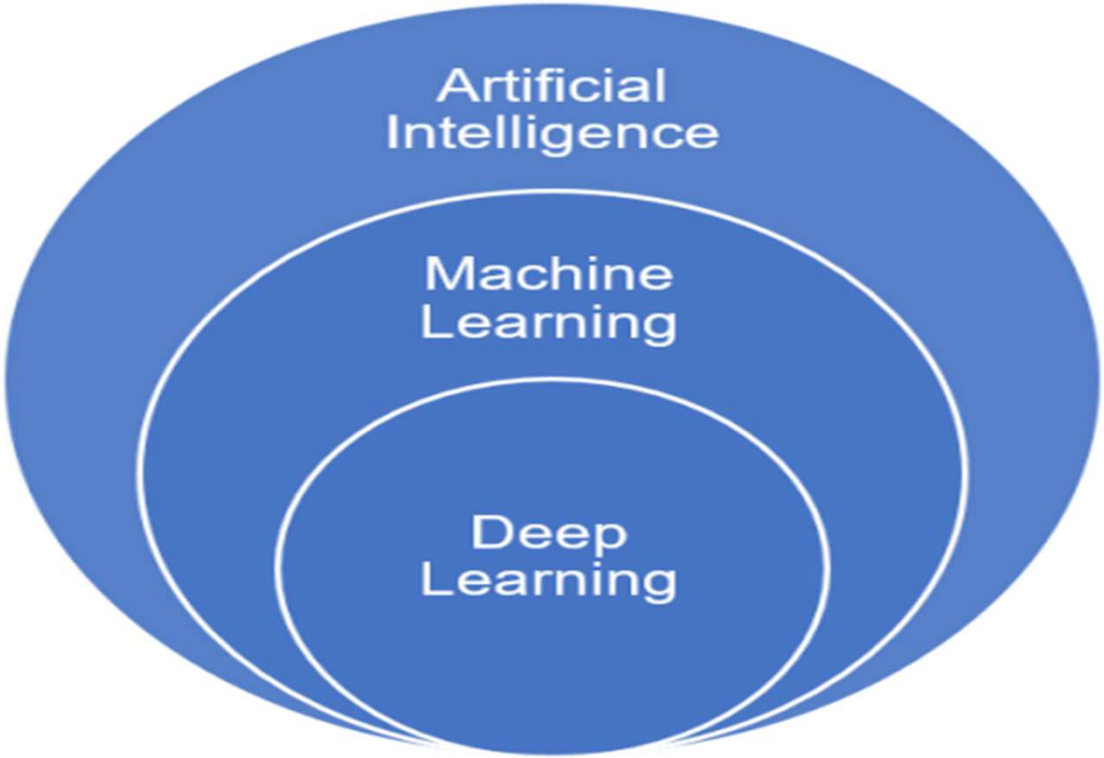
Ability to deal with uncertainties

Learning and Adaptation

Decision Making

- The basis of decision making depends upon the availability of information and how we experience and understand it.
- ‘Information’ includes our past experience, intuition, knowledge, and self-awareness.
- Having knowledge, experience, or insights given a certain situation, helps us visualize what the outcomes could be. and how we can achieve/avoid those outcomes.

AI, ML & DL

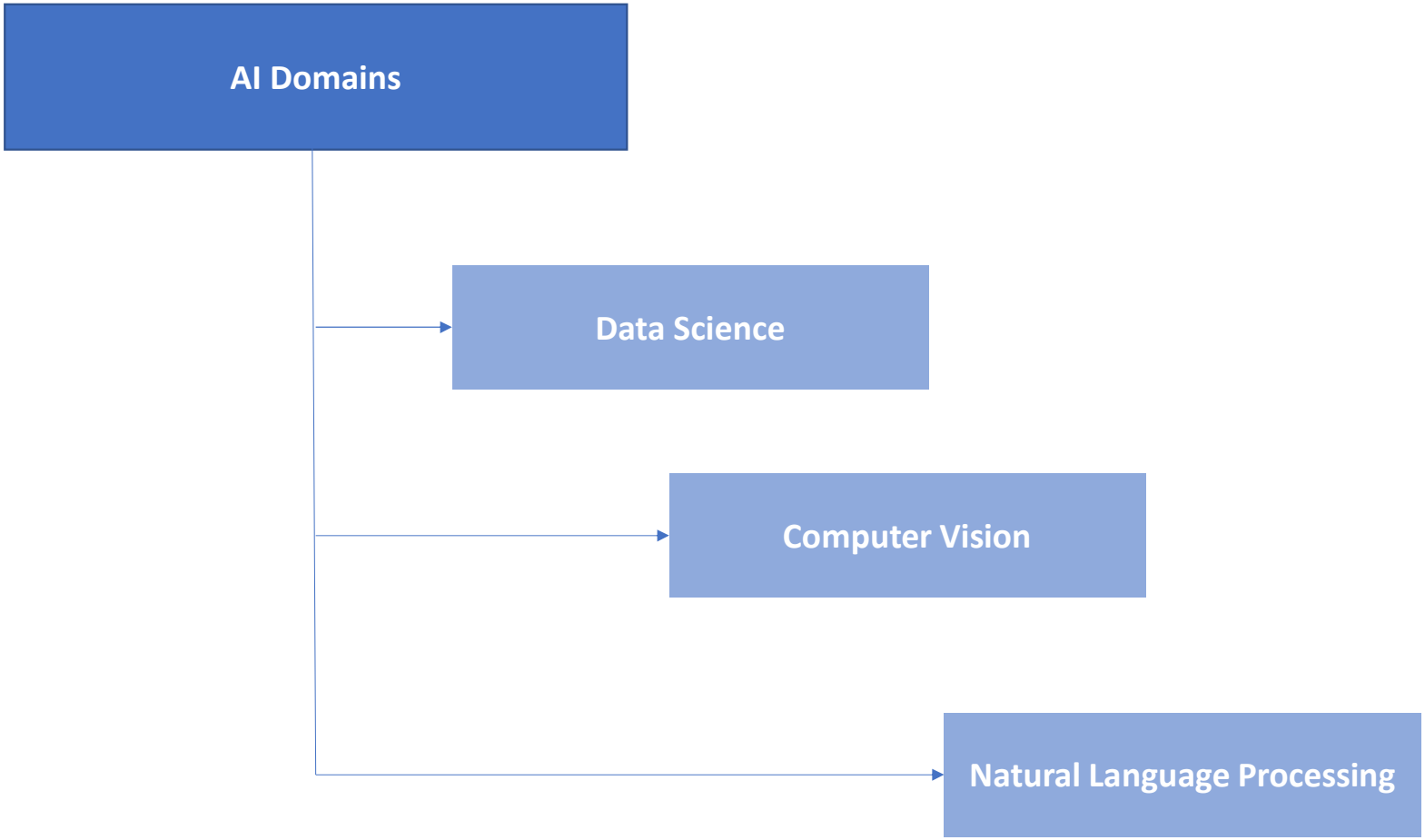


- **Artificial Intelligence (AI)** Refers to any technique that enables computers to mimic human intelligence. It gives the ability to machines to recognize a human's face; to move and manipulate objects; to understand the voice commands by humans, and also do other tasks. The AI-enabled machines think algorithmically and execute what they have been asked for intelligently.

- **Machine Learning (ML)** It is a subset of Artificial Intelligence which enables machines to improve at tasks with experience (data). The intention of Machine Learning is to enable machines to learn by themselves using the provided data and make accurate Predictions/ Decisions.

- **Deep Learning (DL)** It enables software to train itself to perform tasks with vast amounts of data. In Deep Learning, the machine is trained with huge amounts of data which helps it in training itself around the data. Such machines are intelligent enough to develop algorithms for themselves.

There are a lot of applications of AI out of which few are those which come under ML out of which very few can be labelled as DL. Therefore, Machine Learning (ML) and Deep Learning (DL) are part of Artificial Intelligence (AI), but not everything that is Machine learning will be Deep learning


AI Domains
Data Science
Computer Vision
Natural Language Processing

1. Data Science

Data science is a domain of AI related to data systems and processes, in which the system collects numerous data, maintains data sets and derives meaning/sense out of them. The information extracted through data science can be used to make a decision about it.

- Based on strict analytical evidence
- Deals with structured & unstructured data
- Includes various data operations

For examples:-
•**Healthcare:** Data science can identify and predict disease, and personalize healthcare recommendations.
•**Transportation:** Data science can optimize shipping routes in real-time.
•**Sports:** Data science can accurately evaluate athletes' performance.
•**E-commerce:** Data science can automate digital ad placement.
•**Gaming:** Data science can improve online gaming experiences.

2. Computer Vision

Computer Vision(CV) It is a field of science that deals with how computers gain a high level of understanding from digital images or videos. It is a flown that studies how the human visual system works.
The computer vision includes the following methods to produce information:

1. Acquiring Images
2. Processing Images
3. Analysing Images
4. Understanding Images

For examples:-

- **Self-Driving cars/ Automatic Cars:** CV systems scan live objects and analyse them, based on whether the car decides to keep running or to stop.
- **Face Lock in Smartphones:** Smartphones nowadays come with the feature of face locks in which the smartphone's owner can set up his/her face as an unlocking mechanism for it. The front camera detects and captures the face and saves its features during initiation. Next time onwards, whenever the features match, the phone is unlocked

3. Natural Language Processing

Natural Language Processing, abbreviated as NLP, is a branch of artificial intelligence that deals with the interaction between computers and humans using the natural language. Natural language refers to language that is spoken and written by people, and natural language processing (NLP) attempts to extract information from the spoken and written word using algorithms.

For example:

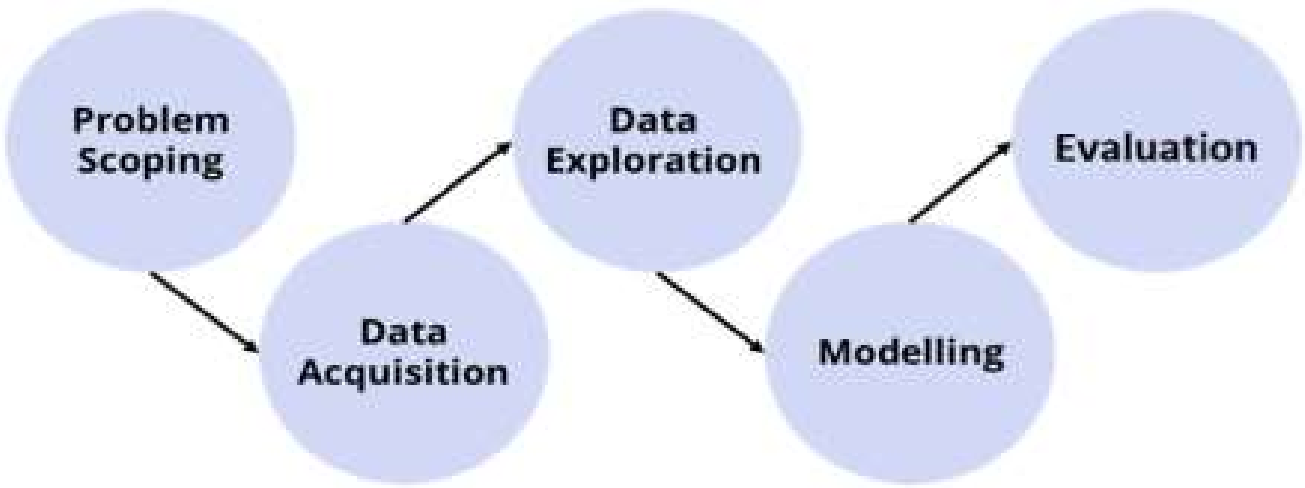
- **Email filters**

Email filters are one of the most basic and initial applications of NLP online. It started out with spam filters, uncovering certain words or phrases that signal a spam message.

- **Smart assistants**

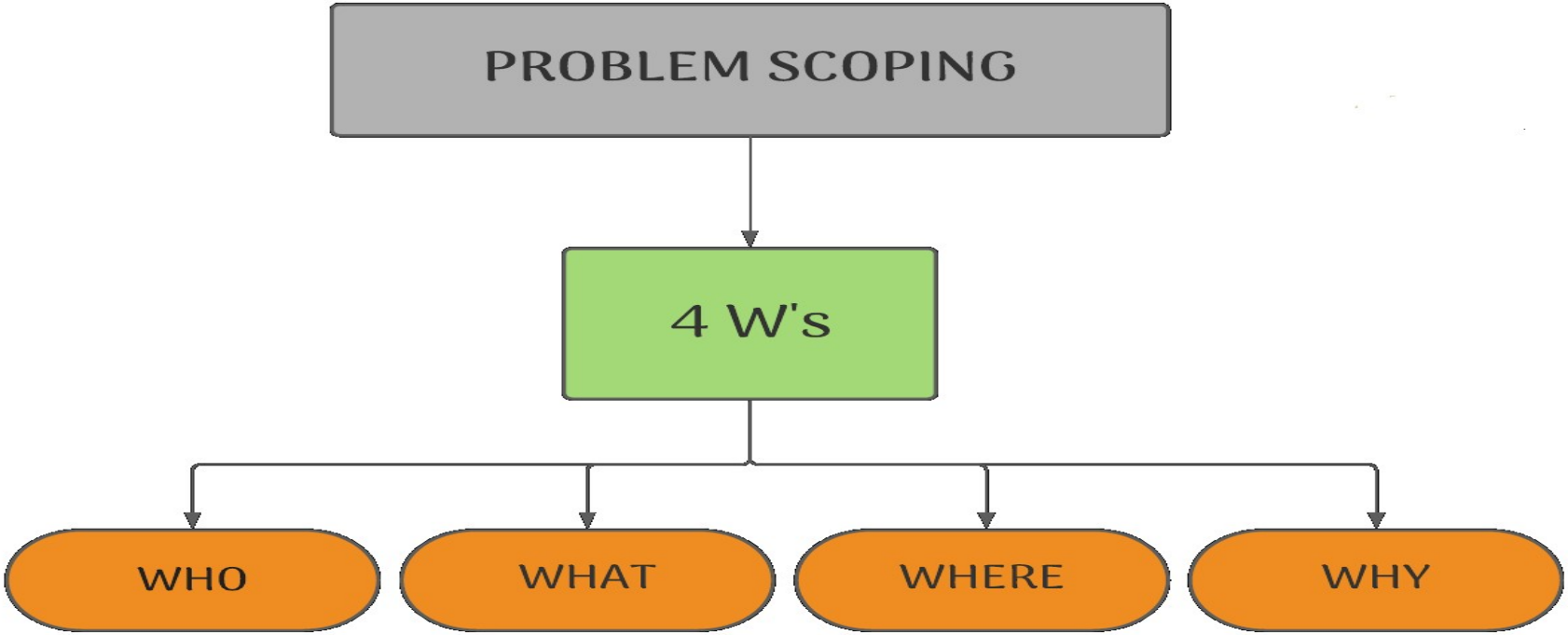
Smart assistants like Apple's Siri and Amazon's Alexa recognize patterns in speech, then infer meaning and provide a useful response.

AI Project Cycle



Problem Scoping

The problem scoping refers to the identification of a problem and the vision to solve it.



The 4Ws of Problem Scoping

1. **Who?** - Refers that who is facing a problem and who are the stakeholders of the problem.
2. **What?** - Refers to what is the problem and how you know about the problem.
3. **Where?** - It is related to the context or situation or location of the problem.
4. **Why?** - Refers to why we need to solve the problem and what are the benefits to the stakeholders after solving the problem

Example of problem scoping:

- Environment
- Agriculture
- Traffic
- Women safety
- Health
- Security

Agriculture example:-

WHO	Stakeholders	Farmers, Fertilizer Producers, Labours, Tractor Companies
WHAT	The problem, Issue, Need	Determine what will a good time for seeding or crop harvesting?
WHERE	Situation	Decide the mature age for the crop and determine its time
WHY	Ideal Solution or Benefit	Take the crop on time and supply against market demand on time

Data Acquisition

Data Acquisition consists of two words:

1. **Data :** Data refers to the raw facts , figures, or piece of facts, or statistics collected for reference or analysis.
2. **Acquisition**: Acquisition refers to acquiring data for the project.

Data Acquisition is the process of collecting accurate and reliable data to work with.The stage of acquiring data from the relevant sources is known as data acquisition.

Data Source

- Survey
- Sensor
- Camera
- Observation

Classification of Data

Data classified as structured, unstructured, or semi-structured data.

1. Structured Data:

•Structured data is generally tabular data that is represented by columns and rows in a database.

•Databases that hold tables in this form are called ***relational databases***.

•The mathematical term "*relation"* specify to a formed set of data held as a table.

•In structured data, all row in a table has the same set of columns.

•SQL (Structured Query Language) programming language used for structured data.

Structured Data:

id	name	age
1	Jim	28
2	Pam	26
3	Michael	42

id	subject	Teacher
1	Languages	John Jones
2	Track	Wally West
3	Swimming	Arthur Curry
4	Computers	Victor Stone

student_id	subject_id	grade
2	1	98
1	2	100
1	4	75
3	3	60
2	4	76
3	2	88

Semi structured Data

- It is the combination of both structured and unstructured data.
- Some data can have a structure like a database whereas some data can have markers and tags to identify the structure of data.
- XML (Extensible Markup Language)data is an example of semi-structured data.

```
<University>
 <ID Number="160801">
  <Name="Shaun">
  <Age="18">
  <Program="Communication">
 <ID Number="160802">
  <Name="Linh">
  <Age="20">
  <Program="Accounting">
..... </University>
```

Unstructured Data

•**Unstructured data** is information that either does not organize in a pre-defined manner or not have a pre-defined data model.
•Unstructured information is a set of text-heavy but may contain data such as numbers, dates, and facts as well.
•**Videos, audio, and binary** data files might not have a specific structure. They're assigned to as **unstructured** data.

Structured Data	Unstructured Data	Semi-structured Data
Well organised data	Not organised at all	Partially organised
It is less flexible and difficult to scale. It is schema dependent.	It is flexible and scalable. It is schema independent.	It is more flexible and simpler to scale than structured data but lesser than unstructured data.
It is based on relational database.	It is based on character and binary data.	It is based on XML
Versioning over tuples,row,tables	Versioning is like as a whole data.	Versioning over tuples is possible.
Easy analysis	Difficult analysis	Difficult analysis compared to structured data but easier when compared to unstructured data.
Financial data, bar codes are some of the examples of structured data.	Media logs, videos, audios are some of the examples of unstructured data.	Tweets organised by hashtags, folder organised by topics are some of the examples of unstructured data.

Dataset

A dataset is a collection of data in which data is arranged in some order. A dataset can contain any data from a series of an array to a database table. Below table shows an example of the dataset:

Country	Age	Salary	Purchased
India	38	48000	No
France	43	45000	Yes
Germany	30	54000	No
France	48	65000	No
Germany	40		Yes
India	35	58000	Yes

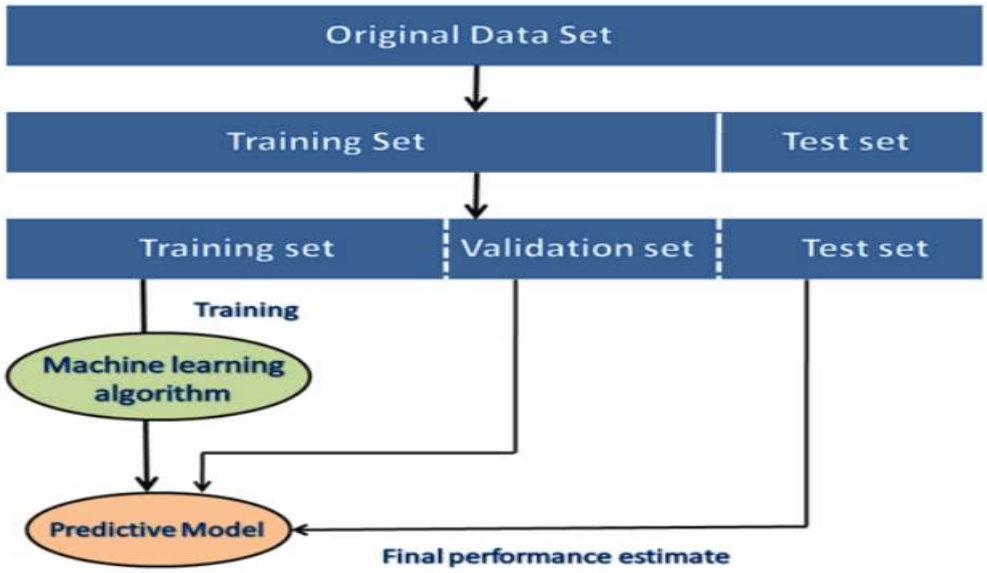
types of data include: text data, image data, audio data, video data, numeric data

Purposes of dataset:

There are 2 key purposes of an AI/ML dataset:
1.To train the model
2.To measure the accuracy of the model once it is trained

Types of Data sets:-

1. Training dataset:
2. Test Dataset



Training dataset

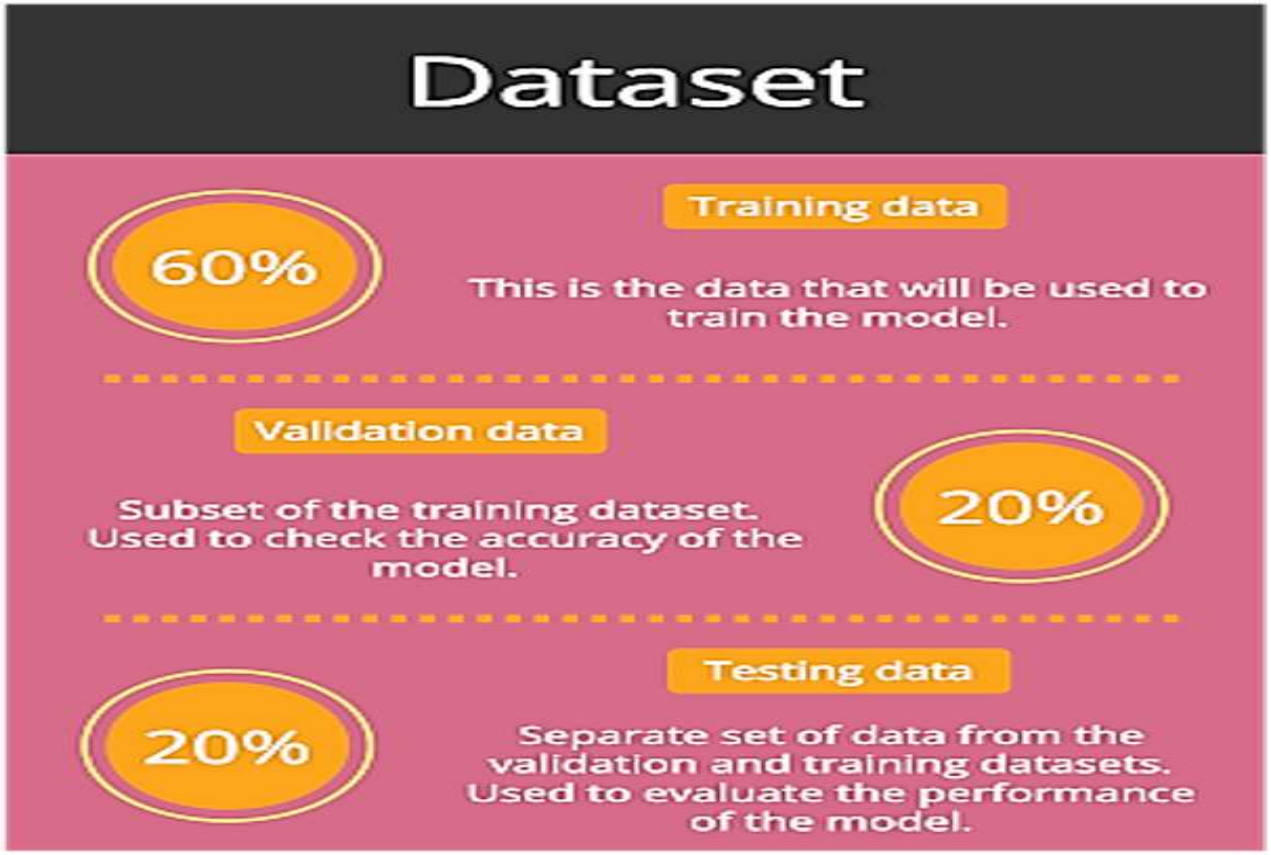
- This is one of the most important subsets of the whole dataset, comprising about 60% of the total dataset.
- This set comprises the data that will initially be used to train the model.
- In other words, it helps teach the algorithm what to look for in the data.
- For instance, a vehicle license plate recognition system will be trained with image data with labels indicating the location and the data format of the license plates of vehicles

Validation dataset

- This subset is about 20% of the total dataset and is used to evaluate all the parameters of the model after the training phase is complete.
- The validation data is known data that helps to avoid training your algorithm on the same type of data and making biased predictions.
- This data is also used to identify if the model is over/underfitting.

Test dataset

- This subset is input at the final stage of the training process and accounts for the last 20% of the dataset.
- The data in this subset is unknown to the model and is used to test the accuracy of the model.
- In simpler words, this dataset will show how much your model has learned from the previous 2 subsets.



Data Collection Methods

- ➢Interviews
- ➢ Questionnaires and surveys
- ➢Observations
- ➢Documents and records
- ➢Focus groups
- ➢Sensors
- ➢Open Public Government Portals
- ➢Reliable Websites (e.g. Kaggle)
- ➢World Organizations' public statistical websites

Data set Source

Open datasets:-Popular open datasets include:
•Google dataset search
•AWS public datasets
•Kaggle datasets
Public government datasets
Popular public government datasets include:
•Data.Gov.uk
•EU open data portal
•Data USA
Image datasets
Popular image datasets include:
•Google's open images
•Coco dataset
•Imagenet
Audio datasets
Popular audio datasets include:
•Environmental audio datasets
•Speech commands dataset
•Free music archive (FMA)

Handling Missing Values

1. Ignore the tuple

➢ This method is not very effective, unless the tuple contains several attributes with missing values.
➢ It is especially poor when the percentage of missing values per attribute varies considerably.

2. Fill in the missing value manually

In general, this approach is time-consuming and may not be feasible given a large data set with many missing values.

3. Use a global constant to fill in the missing value

➢ Replace all missing attribute values by the same constant, such as a label like "Unknown" or $-\infty$.
➢ If missing values are replaced by, say, "Unknown," then the algorithm may mistakenly think that they form an interesting concept, since they all have a value in common—that of "Unknown."
➢ Hence, although this method is simple, it is not foolproof.

4. **Use the attribute mean/most frequent value to fill in the missing value**

➢ For example, suppose that the average age of All passengers on Titanic is 29.7. Use this value to replace the missing value for "Age". (Note: Age is a numeric attribute)

➢ Similarly for "Cabin" if the most frequent value for the "Cabin" is "B96 B98" then use it to replace all the missing values for "Cabin".

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
5	6	0	3	
6	7	0	1	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	
5	Moran, Mr. James	male	NaN	0	
6	McCarthy, Mr. Timothy J	male	54.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S
5	0	330877	8.4583	NaN	Q
6	0	17463	51.8625	E46	S

First 7 rows of Titanic Data

Data Exploration

Data Exploration is the process of arranging the gathered data uniformly for a better understanding. Data can be arranged in the form of a table, plotting a chart, or making a database.

Purpose of Data Exploration:

It is possible that some errors will occur while collecting data. Let us first look at the different types of issues that can arise:

1. **Erroneous Data**: There are two ways in which the data can be erroneous:
    a. **Incorrect values:**
        ➢The values in the dataset are incorrect (at random locations).
        ➢For example, in the phone number column, there is a decimal value, and in the marks column, there is a name mentioned, and so on.
        ➢These are incorrect values that do not correspond to the type of information expected in that position
    b. **Invalid or Null values:**
        ➢At some places, the values get corrupted and hence they become invalid.
        ➢Many times you will find NaN values in the dataset.
        ➢These are null values which do not hold any meaning and are not processible.
        ➢As a result, when these values are encountered, they are removed from the database

2. **Missing Data:**

➢Some cells in some datasets remain empty. Because the values for these cells are missing, the cells remain empty.

➢Missing data cannot be interpreted as an error because the values are not incorrect and may not be missing due to an error.

3**. Outliers:**

➢Outliers are data points that do not fall within the range of a specific element.

➢To better understand this, consider the grades of students in a class. Assume a student missed an exam and thus received no credit for it.

➢If his grades are included, the average for the entire class falls. To avoid this, the average is calculated for the range of marks from highest to lowest, while keeping this specific result separate.

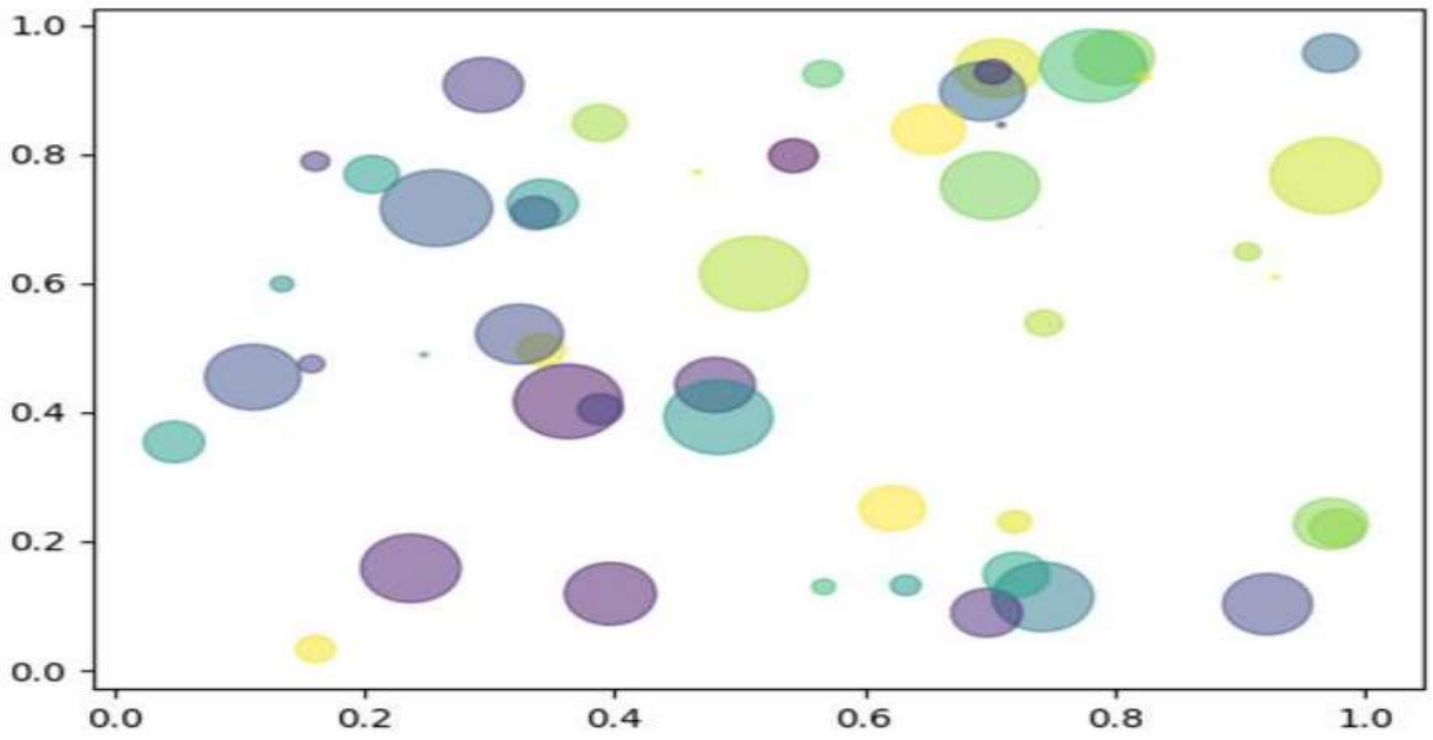
➢This ensures that the class average marks are correct based on the data.

Data Visualization

➢Analyzing the collected data can be challenging because it is all about tables and numbers.
➢While machines can process numbers efficiently, humans require visual aid to understand and comprehend the information passed.
➢As a result, data visualization is used to interpret the collected data and identify patterns and trends.

Scatter plots

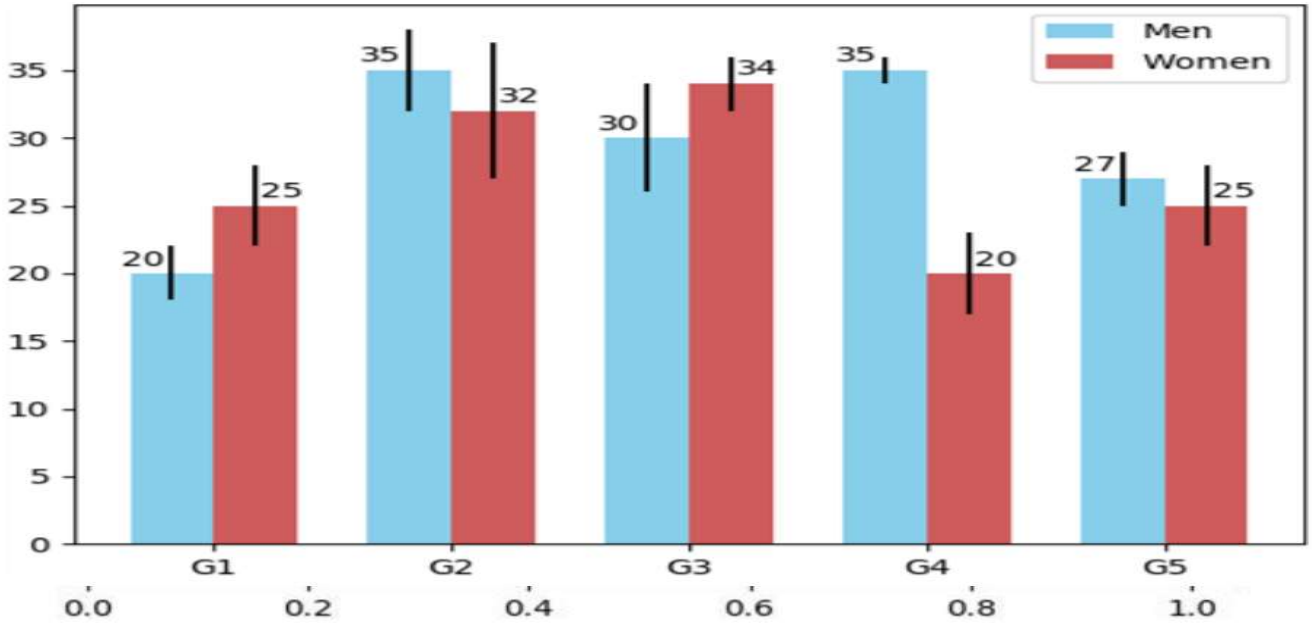
- Scatter plots are used to plot discontinuous data; that is, the data which does not have any continuity in flow is termed as discontinuous. There exist gaps in data which introduce discontinuity.
- A 2D scatter plot can display information maximum upto 4 parameters.



Bar Chart

There are several types of bar charts, such as single bar charts, double bar charts, and so on.

- ➢ The bar chart also works with discontinuous data and is made at uniform intervals.
- ➢ Use bar charts to compare categories when you have at least one categorical or discrete variable.
- ➢ Each bar represents a summary value for one discrete level, where longer bars indicate higher values.
- ➢ Types of summary values include counts, sums, means, and standard deviations.
- ➢ Bar charts, unlike histograms, have spaces between the bars to emphasize that each bar represents a discrete value, whereas histograms are for continuous data



Line Chart

A line chart, also known as a curve chart, is a type of chart that displays information as a series of data points called "markers" that are connected by straight line segments.

➢ It is a basic type of chart that is used in many fields. It's similar to a scatter plot, except the measurement points are ordered (typically by x-axis value) and connected with straight line segments.

➢ A line chart is frequently used to depict a trend in data over time intervals.